

1 Homework

1. Suppose $(X, \mathcal{T})$ is a topological space. Let $f : X \rightarrow \mathbb{R}$ be continuous on $K \subseteq X$ compact. Then $\exists x_1, x_2 \in X$ such that $\forall x \in X$.

$$f(x_1) \leq f(x) \leq f(x_2)$$

2. Show that if a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x) = 0$ for all $x \in \mathbb{Q}$ then $f(x) = 0$ for all $x \in \mathbb{R}$.
3. Give an example of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ and an open set U such that $f(U)$ is not open.

4. Suppose the functions $f_n(x)$ on $\mathbb{R}$ satisfy:

- (a) $0 \leq f_n(x) \leq 1$ for all $x \in \mathbb{R}$ and $n \geq 1$.
 (b) $f_n(x)$ is increasing in x for every $n \geq 1$.
 (c)

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

for each $x \in \mathbb{R}$, where f is continuous on $\mathbb{R}$.

- (d)

$$\lim_{x \rightarrow -\infty} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} f(x) = 1.$$

Show that $f_n(x) \rightarrow f(x)$ uniformly on $\mathbb{R}$.

5. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$|f(x) - f(y)| \leq M|x - y|$$

for all $x, y \in \mathbb{R}$. This type of function is called Lipschitz. Suppose that f is differentiable.

- (a) Show that f is continuous.
 (b) Show that

$$|f'(x)| \leq M$$

for every $x \in \mathbb{R}$.

- (c) Conversely, suppose that

$$|f'(x)| \leq M$$

for all $x \in \mathbb{R}$. Show that

$$|f(x) - f(y)| \leq M|x - y|$$

for all $x, y \in \mathbb{R}$.

2 Discussion

Questions from the Exam 1 Study Guide.